

Software Lab

Team Name : - MedMatrix

Features :

Website Name :- CareGrid.io

1. Organ Donation & Compatibility Matching Engine

Since organ matching is time-sensitive and strict, this feature adds technical depth to your backend and database structure:

- **Compatibility Scoring Algorithm:** Automated matching based on Blood Group, Tissue Typing (HLA compatibility), Urgent Needs Index, and Geolocation/Distance.
- **Recipient Waiting List Priority:** Queue management based on urgency score (e.g., MELD/PELD score) and time on the waiting list.
- **Organ Cold-Ischemia Time Tracker:** Live countdown timers for preserved organs (e.g., Heart ~4 hours, Kidney ~24 hours) with alert thresholds.
- **Anonymous Living Donor Registry:** Secure portal for living donors to register and view eligibility screening checklists.

2. Smart Blood Bank & Emergency SOS

- **Real-time Inventory & Expiration Alerts:** Track blood components separately (Whole Blood, Red Blood Cells, Platelets, Plasma) with automatic notifications when stocks drop low or approach expiration dates.
- **Emergency Broadcast System (SOS):** Automated SMS / Email blast to registered donors of specific blood types when an urgent match is needed in the ICU.
- **Donor Eligibility & Interval Tracker:** Automatically block donors who donated less than 56 days ago to enforce medical safety guidelines.
- **Geofenced Donor Search:** Map UI showing nearby eligible donors relative to the hospital.

3. Daily Patient Update & Doctor Dashboard

- **EHR Timeline & Vitals Logging:** Daily temperature, blood pressure, oxygen levels, and medication charts visualized on line graphs.
- **Attendant / Family Portal:** A read-only mobile/web view where family members can see daily updates written by doctors or nurses without calling ICU desks.
- **Ward & Bed Management:** Visual grid layout of hospital rooms showing Occupied, Available, Cleaning, or Reserved beds.

4. Pharmacy & Prescription Management (E-Prescriptions)

- **Digital Prescription Builder:** Doctors generate prescriptions; automatically sent to the hospital pharmacy.
- **Drug Interaction / Allergy Warning:** Alerts if a prescribed drug interacts dangerously with another active prescription or a listed patient allergy.
- **Automated Stock Deduction:** Dispensing medicine automatically reduces the internal inventory stock.

5. Billing, Insurance & Discharge Workflow

- **Itemized Billing Generator:** Auto-calculates room charges (per day), pharmacy items, surgery fees, and doctor consultation fees.
- **Insurance Claim Status Tracker:** Manage claims submitted to insurance providers with Approval/Rejection tracking.
- **Digital Discharge Summary:** One-click generation of the patient's complete stay report, lab tests, and post-discharge medication schedule as a downloadable PDF.